# ECE 310 Fall 2026

## Lecture 2: Math preliminaries, complex numbers

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# Learning Objectives

- Explain what defines a complex number, be able to identify the magnitude and phase of a complex number, and visualize them on the complex plane.
- Convert between rectangular and polar forms of complex numbers.
- State Euler's identities for complex numbers.
- Interpret and visualize common discrete-time signals including Kronecker delta and unit step.



# Complex Numbers: Rectangular Form

For a complex number $x \in \mathbb{C}$, we may express $x$ in *rectangular form*

$$x = a + jb$$

- $a$:
- $b$:
- $j$:

We may also denote $x^*$ as the *complex conjugate* of $x$ where $x^* = a - jb$.



# Complex Numbers: Polar Form

For a complex number $x \in \mathbb{C}$, we may express $x$ in *polar form*

$$x = Re^{j\theta}$$

- $R$:
- $\theta$:
- $j$:

We may also denote $x^*$ as the *complex conjugate* of $x$ where $x^* = Re^{-j\theta}$.



# Visualizing Complex Numbers



# Magnitude and Phase

Complex numbers are described by two important quantities known as *magnitude* and *phase*. Let $x = a + jb = Re^{j\theta}$

- $\text{Magnitude}(x) = |x| = \sqrt{a^2 + b^2} = \sqrt{xx^*} = R \geq 0$
- $\text{Phase}(x) = \angle x = \begin{cases} \tan^{-1}\left(\frac{b}{a}\right), & a \geq 0 \\ \tan^{-1}\left(\frac{b}{a}\right) + \pi, & a < 0,\ b \geq 0 \\ \tan^{-1}\left(\frac{b}{a}\right) - \pi, & a < 0,\ b < 0 \end{cases} = \theta \in [-\pi, \pi].$

# Magnitude and Phase Examples

Determine the magnitude and phase of the following complex numbers and sketch them on the complex plane.

(a) $x = -4 + j3$

(b) $y = j2$



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Special case to consider: What are the magnitude and phase of $x = -3.10$?

# Complex Arithmetic: Addition and Subtraction

Let $x = a + jb = Re^{j\theta}$ and $y = c + jd = Se^{j\phi}$



# Complex Arithmetic: Multiplication and Divison

Let $x = a + jb = Re^{j\theta}$ and $y = c + jd = Se^{j\phi}$



# Euler's Identities

Euler's formula allows us to convert from polar to rectangular forms for complex numbers. Let $x = Re^{j\theta}$.

$$x = R\cos\theta + Rj\sin\theta$$

This leads to a few important identities:

- $e^{\pm j\pi} = -1$
- $\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}$
- $\sin\theta = \frac{e^{j\theta}-e^{-j\theta}}{2j}$



# Complex Arithmetic Examples

Simplify the following expressions into rectangular and polar forms.

(a) $j2 + \sqrt{2}e^{-j\frac{\pi}{4}}$

(b) $\frac{j3}{1-j}$



Let's consider an example length–7 signal:

$$x[n] = \{1, 4, -2, 0, 3, 1, 2\}$$

Is there something missing here?



Let's consider an example length–7 signal:

$$x[n] = \{1, 4, -2, 0, 3, 1, 2\}$$

Is there something missing here? We don't know where the signal stops and ends!

$$x[n] = \{1, \mathbf{4}, -2, 0, 3, 1, 2\}$$

$$x[n] = \{1, \underset{\uparrow}{4}, -2, 0, 3, 1, 2\}$$

$$x[n] = \{1, \underline{4}, -2, 0, 3, 1, 2\}$$

$$\{x[n]\}_{n=-1}^{5} = \{1, 4, -2, 0, 3, 1, 2\}$$



# Kronecker Delta

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases} .$$

- Also known as unit impulse or unit pulse
- **This is not the same as the Dirac delta!**

Sketch $x[n] = -2\delta[n+3]$

Write $x[n] = \{2, 0, \underset{\uparrow}{3}, -1, -4\}$ in terms of $\delta[n]$.



# Unit Step

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} .$$

- We can flip and shift just like with Kronecker deltas.

Sketch $x[n] = u[3 - n]$



# Sinusoids and Exponentials

- Sinusoids

$$x[n] = A\sin(\omega_0 n + \theta),\ A \in \mathbb{R}, \omega_0 \in \mathbb{R}, \theta \in \mathbb{R}$$

- Exponentials

$$x[n] = Ba^n, a \in \mathbb{C},\ B \in \mathbb{C}$$

